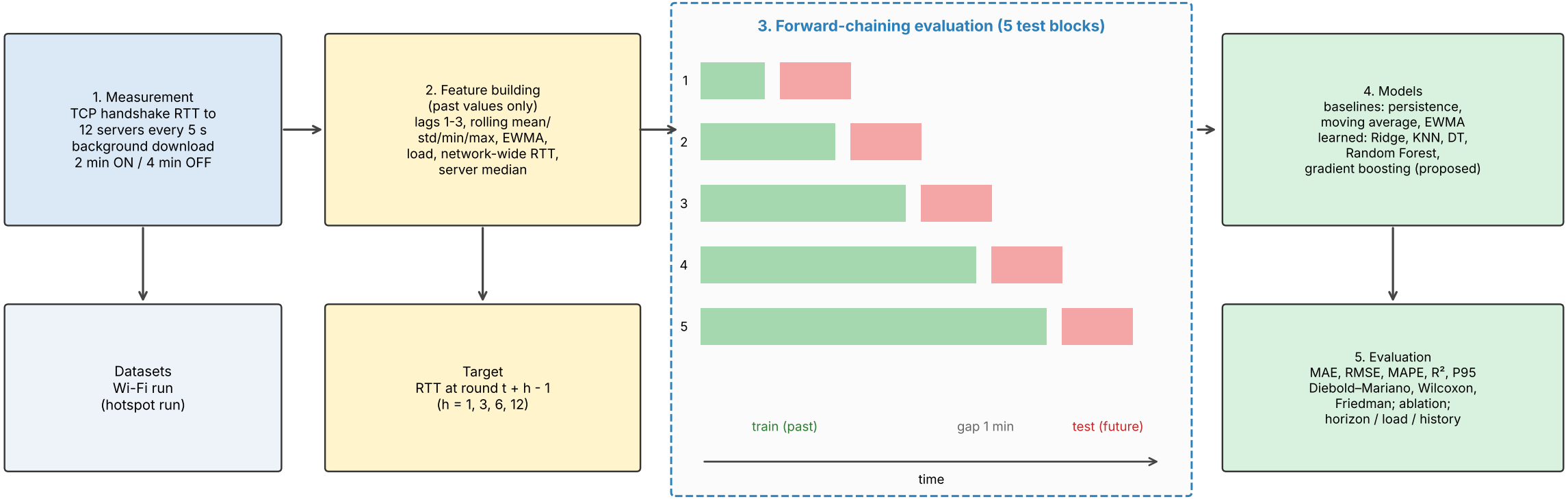
1. Measurement
TCP handshake RTT to
12 servers every 5 s
background download
2 min ON / 4 min OFF
2. Feature building
(past values only)
lags 1-3, rolling mean/
std/min/max, EWMA,
load, network-wide RTT,
server median
3. Forward-chaining evaluation (5 test blocks)
1
2
3
4
5
train (past)
gap 1 min
test (future)
time
4. Models
baselines: persistence,
moving average, EWMA
learned: Ridge, KNN, DT,
Random Forest,
gradient boosting (proposed)
Datasets
Wi-Fi run
(hotspot run)
Target
RTT at round t + h - 1
(h = 1, 3, 6, 12)
5. Evaluation
MAE, RMSE, MAPE, R², P95
Diebold–Mariano, Wilcoxon,
Friedman; ablation;
horizon / load / history